VIDEO GAME CREATION

Lesson 13 - A Hit That Reacts

Semester 1, Week 13 - Basic Animation

- **Objectives:** Sequence a visual reaction (flash) before finalizing an existing game-state change, without altering when score/lives/wave/win-lose logic actually resolves; spawn a fully independent, self-cleaning explosion effect; recognize the three different ways this course's engines pause mid-script (or don't) and why.
- **IGDA Tie-In:** Visual Art - basic animation, juice/feedback fundamentals (introductory); Programming - asynchronous/timed execution, event sequencing (intermediate)
- **Featured Engine (lecture demo):** Scratch, contrasted with GameMaker's Alarm-based workaround
- **Materials:** Demo laptop with all 8 engines verified; Student Handouts, one per engine; students' own Lesson 12 projects (already saved); one new asset (explosion_placeholder.png) - Unreal reuses its existing Cube/Sphere shapes, TIC-80 students hand-draw their own explosion sprite

Agenda

0:00-0:05	Welcome back, Lesson 12 recap
0:05-0:15	A Hit That Reacts: Two New Visual Touches
0:15-0:40	Pausing Mid-Script: Three Engine Families, Three Solutions (live demo in Scratch)
0:40-0:50	An Explosion That Minds Its Own Business
0:50-0:55	Introduce lab exercise, hand out engine-specific handouts
0:55-1:50	LAB: Wire up the flash and explosion, verify game logic is unchanged
1:50-2:00	Exit check / quick share-out - best explosion wins bragging rights

Levels for this lesson's exercise

- Level 1 (Minimal): every hit briefly flashes the enemy and spawns an independent explosion effect; score/lives/wave/win-lose logic all still resolve at exactly the same moments as before.
- Level 2 (Complete): the flash and explosion play correctly on every hit with no stuck enemies or leftover explosions, and game-state timing is provably unaffected.
- Level 3 (Refined): smoother color fades, a scaling/animated explosion, or a multi-frame flicker instead of a flat flash.

See: Lesson 13 Teacher's Script, Lesson 13 Prep Checklist, and the per-engine Student Handouts. This lesson restructures hit-handling code more than most weeks - review your demo engine's Handout closely, and skim the others before class.